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# AI CAREERS FOR WOMEN

## LINKEDIN FELLOWSHIP

# SmartStay AI: Predictive Analytics for Tourism & Hotel Revenue Management

**Project Domain**

Excel

**A Collaborative Initiative**

equipping students with practical skills in AI, Data Analytics, and Business Intelligence


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## Problem Statement

The tourism and hospitality industry generates large volumes of booking data through online reservation systems, travel agencies, and hotel management platforms. This data contains valuable information related to customer behavior, booking patterns, seasonal demand, and revenue performance. However, in many organizations this data remains underutilized because it is often stored in raw formats, contains missing values, duplicates, or inconsistent structures, making it difficult to extract meaningful insights.

Hotels and hospitality businesses rely heavily on accurate demand forecasting, pricing strategies, and occupancy optimization to maintain profitability. Without proper data analysis and visualization tools, managers find it challenging to identify booking trends, peak seasons, customer segments, and revenue drivers. As a result, important decisions related to pricing, marketing strategies, and capacity management may be based on incomplete or inaccurate information.

By leveraging AI tools such as Microsoft Copilot and advanced chatbots, this project not only automates the extraction of insights from 150,000+ booking records but also enables predictive analytics for cancellations, revenue forecasting, and seasonality detection. AI assists in identifying high-risk bookings, optimal pricing, and peak demand periods, transforming raw data into actionable strategies.

This project addresses these challenges by applying data cleaning, analysis, visualization, and AI-assisted insights to a real-world hospitality dataset — the Hotel Booking Demand Dataset. The dataset contains more than 150,000 hotel booking records for both City Hotels and Resort Hotels, representing real operational data from the tourism industry.

The key problem addressed in this project is the lack of structured analysis and visualization of hospitality booking data, which prevents organizations from fully understanding revenue patterns, customer behaviour, and demand trends.

Through the development of a Tourism & Hospitality Booking Dashboard and Revenue

Insights system, this project aims to transform raw booking data into clear, interactive, and actionable insights**.** The dashboard will help stakeholders monitor important metrics such as occupancy rate, average daily rate (ADR), revenue per available room (RevPAR), cancellation trends, and seasonal demand.

This solution will benefit hotel managers, revenue managers, and business analysts by enabling them to make data-driven decisions, optimize pricing strategies, reduce revenue loss due to cancellations, and improve overall business performance in the competitive hospitality industry.

## Objectives

The primary goal of this project is to transform raw hospitality booking data into meaningful insights through data cleaning, analysis, visualization, and AI-assisted reporting. The objectives of this project are as follows:

- To use historical booking and cancellation patterns to predict high-risk periods, likely cancellations, and revenue fluctuations using AI-assisted forecasting.
- To provide prescriptive recommendations, such as suggested pricing adjustments or targeted marketing for high-value customer segments.
- To collect and explore the hotel booking dataset to understand its structure, variables, and potential data quality issues such as missing values or inconsistent formats.
- To clean and preprocess the dataset by removing duplicate records, handling missing values, verifying data types, and standardizing relevant fields to ensure reliable analysis.
- To engineer new analytical variables such as *Total Nights* and *Revenue* to enable meaningful hospitality performance analysis.
- To analyze hospitality performance metrics including occupancy rate, cancellation rate, lead time, customer segmentation, and seasonal booking demand.
- To calculate key hospitality KPIs such as:

1. Average Daily Rate (ADR)
2. Revenue per Available Room (RevPAR)
3. Occupancy Rate
4. Cancellation Rate
5. Repeat Guest Percentage
6. AI-assisted identification of emerging revenue patterns, customer booking behaviour predictions, and suggested interventions for maximizing occupancy and ADR and Microsoft Copilot assists in generating pivot tables for cancellations by season, ADR trends, and occupancy rates, while also suggesting visualizations like heatmaps and trend lines to highlight peak periods and customer segments.

- To build an interactive tourism and hospitality dashboard that visually represents booking patterns, revenue insights, and customer behaviour for stakeholders.
- To integrate AI tools such as Microsoft Copilot or AI chatbots to assist in generating analytical summaries, documentation, and insights based on the dataset.
- To document the complete analytical workflow, including data preprocessing, analysis methods, dashboard creation, and insight generation.
- To upload all project files to a GitHub repository and record a demonstration video explaining the working dashboard and key insights.
- Project-Specific Objective: To provide data-driven recommendations for hotel revenue optimization and demand forecasting by identifying peak booking seasons, customer segments, and revenue-generating room categories.

## Methodology

This project follows a structured five-phase workflow to ensure systematic data analysis, visualization, and reporting.

| S no. | Phase | Description |
|---|---|---|

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|---|---|---|
| **1** | **Data Collection & Exploration** | The dataset used for this project is the Hotel Booking Demand Dataset, which contains more than 150,000 hotel booking records covering both City Hotels and Resort Hotels. The dataset includes information related to booking dates, customer type, stay duration, lead time, and pricing.<br><br>During this phase, the dataset structure was explored to understand:<br><br>• Column names and definitions<br>• Data types (categorical, numerical, binary)<br>• Total number of records<br>• Missing values and inconsistencies<br><br>This exploration helped identify potential areas requiring cleaning and transformation before analysis. |
| **2** | **Data Cleaning & Preprocessing** | To ensure accurate KPI calculation and reliable business insights, several preprocessing steps were performed. |

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| | | • Bookings with ***is_canceled = 1*** were filtered out to avoid revenue distortion and ensure that only confirmed bookings were used for revenue analysis.<br>• Missing values in relevant columns such as children and country were handled appropriately. In numerical columns like children, missing values were replaced with 0 where applicable.<br><br>• A new variable called Total Nights was created: Total Nights = stays_in_weekend_nights + stays_in_week_nights This helps calculate the total duration of customer stays.<br>• Since revenue was not directly provided in the dataset, a new column was created: Revenue = ADR × Total Nights. This calculation allows proper measurement of revenue-related KPIs.<br>• Months were categorized into seasons (Summer, Monsoon, Winter) to analyze demand fluctuations and peak travel periods.<br>• All categorical and numerical variables were verified and corrected where necessary to ensure accurate calculations and analysis.<br>• Duplicate records were checked and removed to maintain data integrity and prevent analytical errors. |

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| **3** | **Analysis & AI Integration** | After preprocessing, analytical techniques were applied to identify booking trends, customer behavior patterns, and revenue drivers.<br><br>Excel formulas, pivot tables, and aggregation methods were used to calculate important hospitality KPIs including:<br><br>• Occupancy Rate<br>• Average Daily Rate (ADR)<br>• Revenue per Available Room (RevPAR)<br>• Cancellation Rate<br>• Repeat Guest Percentage<br>• Monthly Revenue Trends<br>• Seasonal Demand Patterns<br><br>AI tools such as Microsoft Copilot or AI chatbots were used to assist in generating analytical summaries, explaining trends, and supporting documentation of findings and utilize derived variables such as Total Nights, Revenue, and Season to identify patterns, anomalies, and predictive trends, supporting decision-making for hotel revenue optimization. |
| **4** | **Visualization / Output** | An interactive Tourism & Hospitality Booking Dashboard was created to present key insights in a clear and visually engaging format.<br><br>The dashboard includes:<br><br>• KPI cards for major performance indicators<br>• Monthly revenue trend charts<br>• Seasonal demand visualizations |

| | | |
|---|---|---|
| | | • Customer segment analysis<br>• Room revenue contribution charts<br><br>Interactive filters allow users to explore booking trends based on hotel type, season, and customer segment.<br><br>This visualization enables stakeholders to quickly understand hospitality performance and make informed strategic decisions. |
| **5** | **Documentation & Submission** | In the final phase:<br><br>• The complete project report was documented.<br>• A video demonstration was recorded explaining the dashboard, methodology, and insights.<br>• The project deliverables were prepared for submission as part of the AI & Data Analytics Fellowship Program |

**Resources and Timeline**

A. Hardware & Software Requirements

The following hardware and software tools are required to successfully complete the project:

- Laptop or Personal Computer with Intel Core i3 processor or higher Minimum 4 GB RAM (8 GB recommended) for smooth data analysis and dashboard development
- Windows 10 / Windows 11 or macOS

- Microsoft Excel (Microsoft 365 / Excel 2019 or later) – for data cleaning, preprocessing, pivot table analysis, and KPI calculations.
- Power BI Desktop – for building interactive dashboards and visualizing hospitality analytics
- Microsoft Copilot / AI Chatbot (ChatGPT / Google Gemini / Copilot Web) – for assisting with data interpretation, report generation, and analytical summaries.
- Web Browser – Google Chrome or Microsoft Edge (latest version)
- Internet Connection – required for AI tools, dataset download, and online documentation

B. Dataset

The dataset used for this project is the Hotel Booking Demand Dataset, which contains historical booking data from the tourism and hospitality industry.
Dataset Details

- Dataset Name: Hotel Booking Demand Dataset
- Total Records: More than 150,000 booking entries
- Data Type: Structured tabular dataset (CSV format)
- Domain: Tourism & Hospitality Analytics

Dataset Source

Dataset obtained from Kaggle: https://www.kaggle.com/code/muqadsaqudoos/hotel-booking-dataset/input

Dataset Description:- The dataset contains detailed booking information for City Hotels and Resort Hotels, including:

- Booking dates
- Hotel type
- Number of guests
- Length of stay (weekday and weekend nights) • Customer type

- Lead time (days between booking and arrival)
- Average Daily Rate (ADR)
- Cancellation status

This structured dataset enables revenue analysis, demand forecasting, customer segmentation, and hospitality performance measurement.

C. Project Timeline:- The project will be completed in a structured four-week timeline as shown below.

| Phase | Task | Tentative Completion Date |
|---|---|---|
| **1** | Dataset collection, exploration, and understanding | *Week 1* |
| **2** | Data cleaning, preprocessing, and validation | *Week 2* |
| **3** | Data analysis, KPI calculations, and AI tool integration | *Week 2 – 3* |
| **4** | Dashboard development, visualization, and testing | *Week 3* |
| **5** | Report writing, video recording and submission | *Week 4* |

This timeline ensures systematic completion of the project from data preparation to final dashboard development and documentation.

## Conclusion

By the end of this project, a cleaned and validated hotel booking dataset will be available, with all preprocessing steps documented, including handling missing values, removal of duplicates, and creation of derived variables such as Total Nights and Revenue. Key hospitality performance indicators—including Occupancy Rate, Average Daily Rate (ADR), Revenue per Available Room (RevPAR), Cancellation Rate, Booking Lead Time, Repeat Guest Percentage, and Monthly and Seasonal Revenue Trends—will be identified and analysed. An interactive Tourism & Hospitality Booking Dashboard will be developed

to visually present booking patterns, revenue insights, and customer behaviour using charts, KPI cards, and filters. The project will also demonstrate the integration of AI tools such as Microsoft Copilot and AI chatbots to generate analytical insights, summaries, and documentation. A video demonstration will explain the project workflow, dashboard functionality, and key hospitality insights derived from the data. Overall, the project will illustrate how data analytics and AI-assisted tools can transform raw hospitality booking data into meaningful insights, enabling hotel managers and business analysts to make data-driven decisions that improve revenue management, demand forecasting, and customer experience. Through AI-assisted analysis, the project not only transforms raw data into actionable insights but also provides predictive capabilities for future booking trends, cancellations, and revenue patterns, ensuring hotel managers can make proactive, informed decisions rather than relying solely on historical trends.

## Future AI Applications in Hotel Revenue Management

1) Dynamic Pricing Suggestions Based on Predicted Demand
    a) AI can analyse historical booking data, seasonality, lead times, and cancellations to predict periods of high or low demand.
    b) Using these predictions, AI can recommend optimal room prices to maximize revenue without manual trial-and-error adjustments.
2) Chatbot-Based Customer Engagement Insights
    a) AI-powered chatbots can interact with guests to collect preferences, provide personalized recommendations, and address queries.
    b) These insights can feed back into your analytics to improve guest satisfaction and predict repeat bookings.
3) AI-Assisted Marketing Targeting High-Value or Repeat Guests
    a) AI can segment customers based on booking behaviour, stay patterns, and spending.
    b) Hotels can target high-value or repeat customers with personalized offers, discounts, or loyalty programs, increasing revenue and customer retention.

## References

The following resources were used for learning, dataset sourcing, and technical guidance during the project.


- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available
- at: https://www.edunetfoundation.org
- LinkedIn Learning. AI and Data Analytics Courses. 2026. Available at: https://www.linkedin.com/learning
- Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/excel
- Kaggle Inc. Hotel Booking Demand Dataset. Available at: https://www.kaggle.com/datasets